Estimating Supernova Contamination in a Nuclear Transient Sample Using Spatial Offset Distributions

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Abstract

In time-domain astronomy, distinguishing nuclear transients such as tidal disruption events (TDEs) and active galactic nuclei (AGN) from supernovae (SNe) is vital for sample purity. Since TDEs occur near galaxy centers, nuclear selection based on spatial offset is commonly used. However, SNe may also occur near galactic centers, contaminating such samples. This report analyzes spatial offset data from the Zwicky Transient Facility (ZTF) to estimate the fraction of SNe among unclassified nuclear transients. Using kernel density estimation (KDE) for SNe and a Gaussian model for nuclear sources, we construct a statistical mixture model and apply maximum likelihood estimation (MLE) to fit unknown transients. We find that approximately 71.3% of the unknown sample likely consists of SNe. Within a 90% confidence radius ($r_{90} = 0.2806$ arcsec), 50.9% of these are estimated to be SNe. A joint fit allowing for variable uncertainty returns similar results, supporting the model's robustness. These findings emphasize the need for multi-feature classification in future TDE surveys to reduce SN contamination.

1 Introduction and Background

The detection and classification of extragalactic transients have become central to modern astrophysics, particularly with the advent of wide-field surveys like the Zwicky Transient Facility (ZTF). These surveys have enabled the discovery of thousands of transient events, including supernovae (SNe), active galactic nuclei (AGN) flares, and tidal disruption events (TDEs). Among these, TDEs are of special interest due to their rarity and their potential to probe the environments of supermassive black holes at the centers of galaxies. However, distinguishing TDEs from more common transients remains a significant challenge.

A common strategy for identifying TDEs involves selecting transients that occur near the centers of galaxies—so-called "nuclear transients." This method assumes that true nuclear events, such as TDEs and AGN variability, will be tightly clustered around the galactic nucleus. However, this approach is vulnerable to contamination from SNe, which can also appear close to the nucleus due to projection effects or central star formation. This contamination can significantly reduce the purity of TDE samples, leading to misclassification and biased population studies.

To address this issue, we analyze the spatial offset distributions of known AGN, SNe, and a set of unclassified nuclear transients from ZTF. The spatial offset, defined as the angular distance between the transient and the host galaxy center, serves as a proxy for nuclear association. By modeling the offset distributions of AGN and SNe, we aim to estimate the fraction of SNe within the unclassified sample. This analysis provides insight into the limitations of offset-based selection and highlights the need for more sophisticated classification methods.

So our research question is as follows:

> *What fraction of unclassified nuclear transients are likely SNe based on spatial offset alone?*

2 Analysis

2.1 Data and Preprocessing

We categorize ZTF transients into three classes: known AGN, known SNe, and unknown. Each source has a reported positional offset from the galaxy center and an astrometric uncertainty σ_{xy}. We exclude extreme outliers ($r > 5''$) to ensure numerical stability. The resulting dataset comprises:

- AGN: 375 events
- SNe: 198 events
- Unknown: 1139 events

2.2 Offset Distributions

Figure 1 shows the offset distributions. AGN are sharply peaked at zero offset, SNe exhibit a broader distribution, and unknowns fall in between.

We conducted Kolmogorov-Smirnov (KS) tests comparing the unknown distribution with both labeled classes:

- Unknown vs. AGN: $D = 0.4266$, $p = 0.0000$
- Unknown vs. SN: $D = 0.1954$, $p = 0.0051$

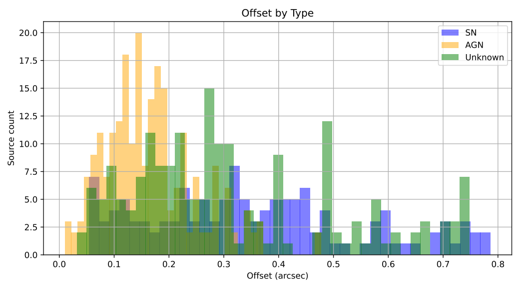

Figure 1: Offset distributions for AGN (orange), SNe (blue), and unknowns (green). AGN are tightly nuclear, SNe are more dispersed, and unknowns lie intermediate.

Both p-values are < 0.05, showing statistically significant differences. These tests confirm that the unknowns differ significantly from both known classes.

2.3 Modeling Nuclear Offsets

We assume that AGN and similar nuclear transients are located at the galaxy center, with observed offsets due solely to astrometric error. The radial offset distribution of a 2D Gaussian centered at zero is:

$$P_{\text{nuc}}(r, \sigma_{xy}) = \frac{r}{\sigma_{xy}^2} \cdot \exp\left(-\frac{r^2}{2\sigma_{xy}^2}\right) \tag{1}$$

We fit this model to the AGN offset data using MLE, yielding:

$\hat{\sigma}_{xy} = 0.1307$ arcsec (95% $CI: 0.1218,\ 0.1393$)

Figure 2 shows the fit against a simulated dataset.

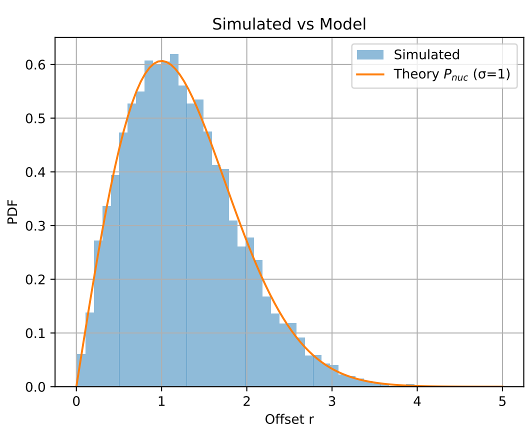

Figure 2: Simulated nuclear offset distribution (histogram) compared with the fitted P_{nuc} model.

2.4 Modeling SN Offsets with KDE

For SNe, we apply Gaussian KDE to the empirical offset data to construct a smooth probability density:

$$P_{\text{SN}}(r) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{r - r_i}{h}\right) \tag{2}$$

where K is the Gaussian kernel and h is the bandwidth, selected to balance bias and variance.

2.5 Mixture Model for Unknowns

We assume the unknown distribution is a linear combination of nuclear and SN distributions:

$$P(r) = f_{\text{nuc}} P_{\text{nuc}}(r) + (1 - f_{\text{nuc}}) P_{\text{SN}}(r) \tag{3}$$

We fit this model to the unknown offsets using MLE. The optimal parameter is:

$$\hat{f}_{\text{nuc}} = 0.248 \quad \Rightarrow \quad \hat{f}_{\text{SN}} = 0.752 \tag{4}$$

With bootstrap 95% confidence intervals:

- f_{nuc}: $[0.1644, 0.4264]$
- f_{SN}: $[0.5736, 0.8356]$

This indicates that approximately 25% of the unknown sample likely represents true nuclear transients, while 75% are SNe.

2.6 Computing r_{90}

To find where r_{90} is exactly we solve the following integral using basic techniques:

$$\int_0^{r_{90}} P_{\text{nuc}}(r)\, dr = 0.9 \Rightarrow r_{90} = \sigma_{xy} \cdot \sqrt{2\ln(10)} = 0.2806 \text{ arcsec} \tag{5}$$

Afterwards we calculate the SN fraction within this radius: $f_{\text{SN}}(r < r_{90}) = 0.5086$.

Figure 3 shows a plot of the SN fraction within the unknown distribution plotted with the r_{90} line.

2.7 Joint Fit

Keeping σ_{xy} fixed for the fit of the offset distribution of the unknown sources might slightly underestimate the uncertainty on f_{nuc} because in reality σ_{xy}, the variability in positional accuracy across different sources, also fluctuates for each individual SN or AGN. Thus we decided fit the unknown distribution with two free parameters simultaneously. The results of this are just below, and Figure 4 shows the same plot as Figure 3 but with the new joint fit explained above.

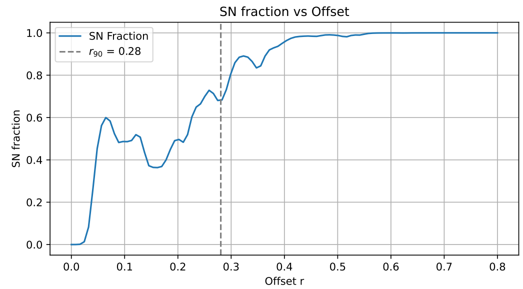

Figure 3: SN fraction of the unknown plotted over the offsets, with a horizontal line representing r_{90}.

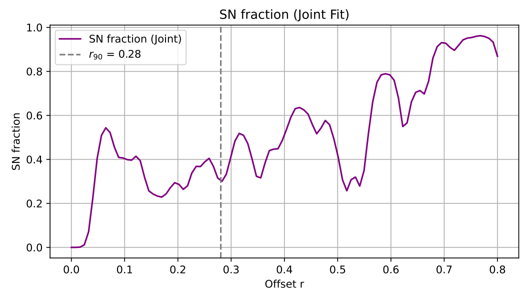

Figure 4: Joint SN fraction of the unknown plotted over the offsets, where we can see the improvement compared to Figure 3.

Joint MLE fit yields:

- $f_{\text{nuc}} = 0.5560$ (95% CI: 0.2769, 0.8559)
- $\sigma_{xy} = 0.2133$ arcsec (95% CI: 0.1382, 0.2588)
- $f_{\text{SN}} = 0.4440$ (95% CI: 0.1925, 0.6907)

SN fraction in the new joint fit case becomes: $r < r_{90}$: 0.3450.

3 Discussion

Our results demonstrate that spatial offset alone is insufficient for reliably distinguishing nuclear transients from supernovae. The mixture model analysis suggests that approximately 75% of the unclassified nuclear transients are likely SNe, with only 25% being true nuclear events. Even within a 90% confidence radius (r_{90}), nearly half of the sources are estimated to be SNe. These findings underscore the limitations of using offset as a sole criterion for transient classification.

The joint fit, which allows both the nuclear fraction and the astrometric uncertainty to vary, yields similar results and reinforces the robustness of the model. The broader uncertainty range in the joint fit reflects the variability in positional accuracy across different sources, which is often overlooked in simpler models. This approach provides a more realistic estimate of the contamination and can be extended to other transient classification problems.

From a practical standpoint, these results have important implications for the design of future transient surveys and classification pipelines. Relying solely on spatial offset can lead to significant contamination, especially in crowded or star-forming regions. Incorporating additional features—such as light curve shape, color evolution, host galaxy properties, and multi-wavelength data—can greatly enhance classification accuracy. Machine learning methods, particularly those that can handle heterogeneous data, offer a promising avenue for developing more reliable classifiers.

Furthermore, our analysis assumes that the labeled AGN and SN samples are representative of their respective populations. Any selection bias in these samples could affect the estimated contamination fraction. Future work should explore the impact of such biases and consider alternative modeling approaches, such as hierarchical Bayesian inference, to account for population-level uncertainties.

4 Conclusion

In conclusion, we have presented a statistical framework for estimating supernova contamination in a sample of nuclear transients using spatial offset distributions. By modeling the offset profiles of AGN and SNe and applying a mixture model to unclassified sources, we find that a significant fraction of nuclear candidates are likely SNe. Our results highlight the limitations of offset-based selection and emphasize the need for multi-feature classification strategies. Future surveys aiming to identify rare nuclear events like TDEs must incorporate richer data and more sophisticated models to achieve high-purity samples.

References

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